



Asymmetric responses to resource overload: a bow-tie perspective on lake eutrophication and human obesity

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Abstract

Biological systems under chronic resource overload often exhibit asymmetric transitions into high-load states that are difficult to reverse. Within a bow-tie (hourglass) framework, such dynamics arise when diverse inputs are funnelled through a constrained regulatory core governing system-level responses. Here, lake eutrophication and human obesity are analysed as structurally distinct yet dynamically analogous manifestations of resource overload. In lakes, external nutrient inputs and internal biogeochemical feedbacks drive shifts from clear-water, macrophyte-dominated regimes to hypertrophic, phytoplankton-dominated states. In humans, sustained caloric surplus interacts with metabolic–hormonal regulation and behavioural–social drivers to promote the development and stabilisation of obesity. In both systems, reinforcing feedbacks organise around a central regulatory core, reducing flexibility, generating hysteresis, and constraining recovery trajectories. Despite these similarities, key asymmetries emerge. In lakes, dynamics under overload are dominated by a limited set of reinforcing feedbacks, whereas in obesity regulation remains distributed across interacting physiological, behavioural, and environmental domains. In both cases, responses involve cascade-like propagation of effects, taking the form of trophic cascades in lakes and cross-domain feedback cascades in humans. These results show that similar system-level dynamics, including alternative stable states, tipping points, and hysteresis with constrained reversibility, can arise from differently structured regulatory architectures. The comparison demonstrates that reversibility is system-specific and shaped by the organisation of the regulatory core and associated feedbacks. Interpreting eutrophication and obesity through a bow-tie framework provides a comparative, architecture-based perspective on resource overload and helps explain why effective interventions require coordinated actions targeting multiple components of the feedback structure.

Keywords Feedback loops · Hysteresis · Alternative stable states · Set-point theory · Complex systems · Energy balance

Introduction

In contemporary science, increasing specialization limits the integration of knowledge across disciplines and hinders the identification of general patterns in complex biological systems (Holling 2001). Conceptual comparisons address this limitation not by asserting equivalence between systems, but by revealing how different regulatory structures can generate similar dynamical behaviours (Olson et al. 2019).

Cross-system approaches are increasingly used in complex systems research, yet the explicit application of bow-tie regulatory architecture as a comparative lens remains rare (Csete and Doyle 2004; Kitano 2004; Friedlander et al. 2015). This study applies this framework by comparing two systems exposed to persistent resource overload: lake ecosystems undergoing eutrophication and hypertrophy, and the human body experiencing overweight and obesity.

Although these systems differ fundamentally in biological organization, both exhibit transitions to altered states driven by feedback-mediated processes and often associated with alternative stable states and hysteresis (Scheffer and Carpenter 2003; Beisner et al. 2003). In lakes, nutrient enrichment stimulates primary production and can lead to hypertrophic conditions characterized by excessive accumulation of organic matter and a restructuring of food

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web interactions (Harper 1992; Henderson 2015; Kornijów 2024). In humans, sustained caloric surplus promotes adipose tissue accumulation and the development of obesity, mediated by metabolic, hormonal, and behavioural mechanisms (Wilborn et al. 2005; Hall 2023; Sorensen 2023). In both cases, system functioning becomes progressively restricted as resource excess alters regulatory processes; however, the underlying mechanisms and the extent of hysteresis differ, reflecting differences in regulatory organization.

Bow-tie architecture provides a unifying framework for analysing these differences. In such systems, diverse inputs are funnelled through a constrained regulatory core that generates multiple outputs (Csete and Doyle 2004; Kitano 2004). This structure supports robust functioning under moderate conditions but creates vulnerability under sustained overload: processing capacity becomes saturated, control concentrates within a limited set of pathways, and reinforcing feedbacks stabilise high-load states and limit recovery trajectories. Existing environmental management frameworks such as DPSIR and DAPSI(W)R(M) successfully organise pressures, states, and responses (Elliott et al. 2017), but are predominantly applied within single domains, and do not explicitly represent the internal regulatory cores, defined here as constrained sets of processes that integrate diverse inputs and generate system-level responses.

Here, the bow-tie architecture provides a conceptual framework for organising known drivers, feedbacks, and state transitions, without relying on explicit network reconstruction. Eutrophication and obesity are analysed as system-specific manifestations of resource overload acting on central regulatory cores. The aim is not to equate lakes with humans, but to show that similar system-level dynamics, including alternative stable states, tipping-point behaviour, and hysteresis with constrained reversibility, can arise from differently organised regulatory architectures. Asymmetries in reversibility reflect differences in regulatory organisation.

Conceptual background

The analysis draws on core concepts from complex systems research, focusing on how regulatory organization shapes system dynamics rather than on specific mechanisms. Feedbacks are processes in which system outputs influence subsequent dynamics, either reinforcing or counteracting change (Holling 1973). Within this framework, resilience denotes the capacity to maintain functional state under perturbation, while hysteresis describes path-dependent responses in which recovery requires different conditions than those that triggered system change (Beisner et al. 2003; Scheffer et al. 2009).

Regulatory architecture refers to the organization of interacting components and feedbacks that delimit the range of attainable system states. It is conceptualized here as a bow-tie structure in which diverse inputs are funnelled through a constrained regulatory core and give rise to multiple outputs (Csete and Doyle 2004; Kitano 2004). In simple terms, bow-tie architectures can be understood as “many inputs – a constrained core – many outputs”.

This organization entails a trade-off between robustness and vulnerability. Concentrating regulation in a core can stabilize system functioning under moderate conditions, but sustained overload can saturate processing capacity, channel dynamics into a limited set of reinforcing feedbacks, and reduce alternative regulatory pathways. Under such conditions, systems may shift into high-load states stabilized by positive feedbacks and characterized by hysteresis.

Here, the bow-tie concept is used at the level of regulatory architecture rather than specific mechanisms to characterize the organization of mechanisms around a central regulatory layer. The focus is not on process equivalence, but on shared architectural properties: a narrow core that stabilizes normal functioning yet, under chronic overload, promotes feedback-dominated states with hysteresis and limited reversibility.

Lake and human body as ecosystems

Lakes are classical examples of ecosystems, consisting of interacting biological communities and their physical environment within a defined spatial unit, shaped by catchment-driven inputs of water and nutrients (Harper 1992; Henderson 2015).

More recently, the ecosystem concept has been extended to the human body, which comprises multiple heterogeneous habitats differing in physical and nutritional conditions and hosting diverse microbial communities (Robinson et al. 2010; Zhou et al. 2013). In this context, the gut microbiome has been implicated in processes associated with obesity (Van de Guchte et al. 2020; Wilson et al. 2020).

The human immediate environment (e.g. urban or rural setting) can be interpreted as a functional analogue of a lake catchment, shaping resource inputs and system dynamics. This analogy is structural and functional rather than spatial and does not imply normative, medical, or evaluative equivalence. This perspective enables comparison of both systems as integrated ecological entities in which structure, function, and regulation emerge across multiple scales, consistent with broader uses of “health” as a descriptor of functional condition in both ecological and biological systems (Sfara and El-Hani 2023). On this basis, both lakes and human bodies can be treated as ecosystems subject to

Table 1 Classification of lake trophic status and human body mass. Alignment is heuristic and does not imply equivalence; drivers differ between systems (external nutrient legacies in lakes vs. interactions between external inputs and internal regulation in humans)

Lake trophic status TSI		Human body mass BMI	
Oligotrophy	≤ 30	Underweight	≤ 18.5
Mesotrophy	30–50	Normal weight	19–24.9
Eutrophy	50–70	Overweight	25–29.9
Hypertrophy	> 70	Obesity	> 30

Table 2 Causes of eutrophication in lake ecosystems and drivers of overweight and obesity in humans

Lake	Human
1. External nutrient inputs: catchment deforestation; agricultural and urban runoff; inflow of sewage; leaky septic systems; municipal and industrial wastewater; atmospheric deposition	1. Excess energy intake and socio-environmental drivers: high consumption of sugars and fats; low fibre intake; highly processed and palatable foods (obesogens); low socioeconomic status; chronic stress; weight stigma; use of certain medications (e.g. psychoactive and antidepressant drugs)
2. Loss of biological regulation: destruction of aquatic vegetation (grazing, mechanical removal, chemical treatments); decline in piscivorous fish (overfishing, winterkill)	2. Physiological and pathological factors: metabolic disorders (e.g. diabetes, dyslipidaemia, gout); hormonal imbalances; potential viral factors (e.g. adenovirus Ad-36)
3. Internal feedback mechanisms: periodic or persistent oxygen depletion; phosphorus release from sediments and loss of oxygen-dependent species	3. Behavioural and lifestyle factors: physical inactivity; sleep disorders (e.g. night eating syndrome); dysregulated behaviour

resource overload, allowing their regulatory architectures to be compared within a common framework.

Indices and classifications

Commonly used indices reduce complex systems to one-dimensional representations of state. In lakes, trophic status is assessed using the Carlson Trophic State Index (TSI; Carlson 1977), whereas in humans body mass is represented by the body mass index (BMI; Burkhauser and Cawley 2008). In both cases, system condition is expressed along a deficiency–excess gradient, and Table 1 provides a heuristic alignment of trophic categories with body weight classes despite fundamental differences in system organisation.

However, such indices compress multidimensional regulatory processes into single metrics, obscuring differences in regulatory architecture and providing limited insight into system dynamics, feedback structure, and responses to perturbation. In particular, they mask the stability landscape and the feedbacks underlying hysteresis, motivating a shift towards an explicit analysis of regulatory architecture.

Regulatory organization and feedback dynamics

In complex systems, reinforcing feedbacks can generate self-amplifying dynamics that drive systems towards high-load states. Such processes are well documented in both hypertrophication (Harper 1992; Scheffer et al. 1993) and obesity (Wilborn et al. 2005; Hall 2023).

These dynamics involve both bottom-up and top-down regulation, defined differently across systems. In lake ecosystems, bottom-up processes are driven by resource availability (e.g. nutrients), whereas top-down regulation reflects trophic control by higher levels (Moss et al. 1994; Stephen et al. 1998). In humans, bottom-up processes include metabolic, hormonal, and genetic regulation, while top-down processes encompass behavioural, social, and environmental influences, including broader institutional and cultural drivers such as public-health policies and social norms (Pietiläinen et al. 2008; Luo and Lin 2021).

Despite these parallels, the organization of feedbacks differs between systems and can be described within a bow-tie architecture, in which diverse inputs are processed through a central regulatory core (Csete and Doyle 2004). In lakes, this core is primarily biogeochemical; in humans, it is physiological. Under sustained overload, dynamics may become dominated by a limited set of reinforcing feedbacks, with alternative regulatory pathways progressively reduced.

Bow-tie architectures stabilise system functioning under moderate conditions but, under sustained overload, promote feedback-dominated dynamics and regime shifts. This behaviour aligns with resilience theory (Holling 1973; Beisner et al. 2003), in which systems may cross critical thresholds and shift into alternative stable states. Differences in the nature of the core influence persistence and recovery potential, and the following sections examine these system-specific mechanisms in detail.

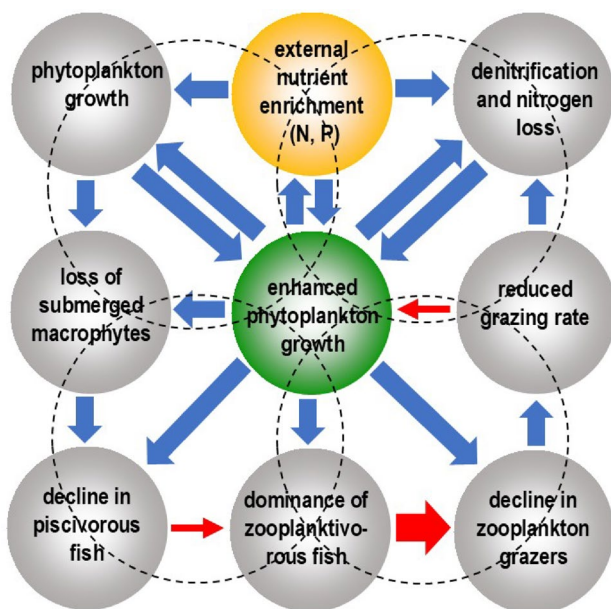
Eutrophication and the state of hypertrophy

The ecological consequences of eutrophication, including algal blooms and oxygen depletion, have been recognised since early limnological studies (Hasler 1947). Elevated trophic status in aquatic ecosystems is primarily driven by increased nutrient inputs from the watershed (Harper 1992; Henderson 2015; Kornijów 2024). Although nutrient enrichment may occur under natural conditions, anthropogenic inputs, such as deforestation, wastewater discharge, and fertiliser use, are the dominant drivers of eutrophication (Table 2).

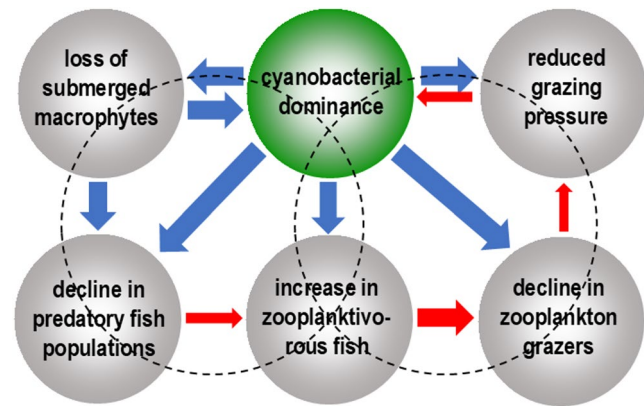
The magnitude and trajectory of ecosystem response depend not only on external inputs but also on internal

Table 3 Structural determinants of system responses to resource overload in lakes and humans

Lake	Human
1. Eutrophication-prone environment: catchment (soil and rock fertility; topography; terrestrial and riparian vegetation); lake morphometry; hydrological regime; climatic variability	1. Obesogenic environment: policy, physical, economic, and cultural conditions shaping food availability and lifestyle
2. Internal physico-chemical regulation: water and sediment properties; phosphorus retention and release regulated by redox conditions	2. Biological predisposition and internal regulation: genetic background; metabolic and hormonal regulation; gut microbiota
3. Biotic structure and interactions: composition; diversity; functional organization of communities	3. Behavioural and socio-cultural factors: dietary patterns; physical activity; social interactions; cultural practices

**Fig. 1** Eutrophication as a coupled bottom-up and top-down system, illustrating interactions among external nutrient inputs, internal biogeochemical processes, and trophic regulation. Resource nodes drive bottom-up pathways (blue), consumer nodes drive top-down forces (red). Dashed lines denote reinforcing feedback loops that organise these interactions around a central regime of phytoplankton dominance, thereby stabilizing the eutrophic–hypertrophic state

system properties, including morphometry, physico-chemical conditions, and biological community structure (Table 3). Increased nutrient availability enhances primary production, particularly phytoplankton growth (Fig. 1). Reduced light penetration leads to the decline and eventual loss of submerged macrophytes, while enhanced primary production increases organic matter loading, intensifying oxygen depletion during decomposition (Phillips et al. 2016; Cheng et al. 2023). Under hypoxic conditions, phosphorus is released from sediments, reinforcing nutrient availability

**Fig. 2** Hypertrophic stage of eutrophication, illustrating the reorganization of the system under sustained resource overload. Blue arrows indicate bottom-up pathways initiated by nutrient enrichment and propagating through cyanobacterial producers (from resources to consumers), whereas red arrows represent top-down forces (from consumers to their resources). Dashed lines denote reinforcing feedback loops. Compared to the more distributed structure shown in Fig. 1, system dynamics become dominated by a limited set of reinforcing feedbacks, reflecting reduced regulatory diversity and a concentration of control within a constrained regulatory core

and further promoting phytoplankton dominance (Søndergaard et al. 2013).

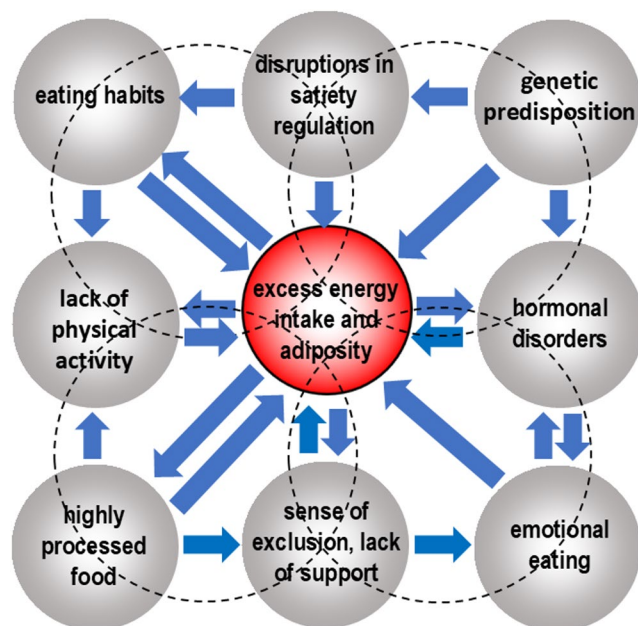
At advanced stages, bottom-up processes interact with top-down mechanisms (Fig. 2). The decline of predatory fish weakens top-down control, allowing zooplanktivorous fish to increase, thereby reducing zooplankton grazing on phytoplankton and amplifying algal blooms (Stephen et al. 1998). This transition reflects a structural reorganisation in which distributed interactions are replaced by a limited set of dominant feedbacks, including trophic cascade effects that amplify phytoplankton dominance (Carpenter et al. 1985). Under hypertrophic conditions, system dynamics become dominated by a small set of reinforcing feedbacks, while alternative regulatory pathways are reduced.

This shift reflects a reduction in regulatory diversity, with control concentrated within a constrained regulatory core, consistent with bow-tie organization. These coupled feedbacks stabilise a hypertrophic state characterised by persistent cyanobacterial dominance. Consequently, cyanobacterial blooms, oxygen depletion, hydrogen sulphide formation, and toxin production (e.g. microcystins) degrade habitat quality and pose risks to aquatic organisms and humans (Barica 1980; Brooks et al. 2016; Le Moal et al. 2019).

Consequently, biodiversity, ecosystem functioning, and service provision are substantially reduced (Table 4). In hypertrophic systems, feedbacks become so strong that additional nutrient inputs produce limited further structural change but accelerate internal accumulation processes.

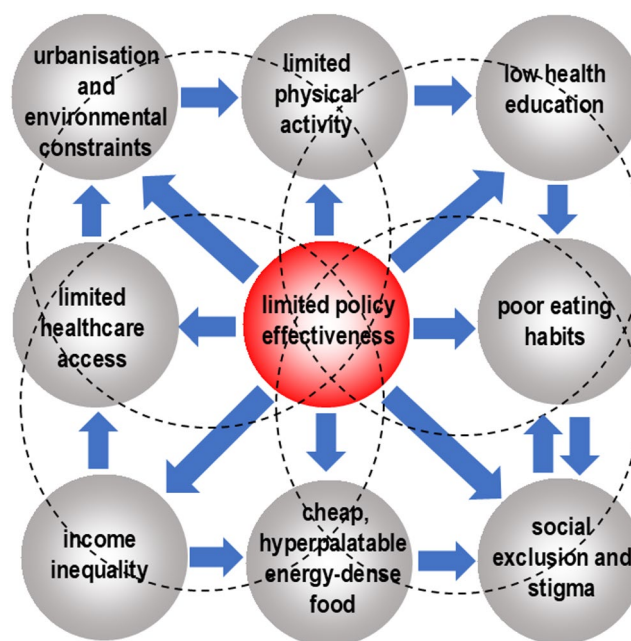
Table 4 Consequences of lake hypertrophy and human obesity

Lake	Human
1. Physiological and metabolic disruption: oxygen depletion; impaired metabolic processes; toxic conditions (microcystins, hydrogen sulphide)	1. Physiological and metabolic impairment: metabolic dysregulation; excessive fat accumulation; reduced adipose tissue oxygenation
2. Loss of biological integrity: elimination of sensitive species; reduction in biodiversity	2. Health deterioration: reduced physical fitness; musculoskeletal disorders; cardiovascular diseases; increased cancer risk
3. Imbalance of system processes: disruption of production–decomposition balance; organic matter accumulation in sediments	3. Functional and phenotypic effects: deterioration of physical condition and appearance; premature system aging (senescence)
4. Decline in system-level functions: deterioration of water quality; reduction of ecosystem services	4. Psychosocial consequences: reduced quality of life; social stigma; social exclusion

**Fig. 3** Bottom-up domain of obesity at the individual level, illustrating feedbacks among metabolic, behavioural, and physiological processes. Blue arrows indicate major positive causal influences among components, and dashed lines denote reinforcing feedback loops. These interacting processes form a structured network organised around a central state (excess energy intake and adiposity), reinforcing conditions that promote further accumulation and persistence of obesity

Weight gain and obesity

Processes underlying obesity follow a sequence analogous to that observed in eutrophication, linking resource input, system response, feedback amplification, and system-level consequences. Obesity develops under sustained positive energy balance, where energy intake chronically exceeds physiological requirements (Wilborn et al. 2005; Hall 2023;

**Fig. 4** Top-down domain of obesity, illustrating how social, economic, and policy-level determinants shape obesogenic environments and behaviours. Blue arrows indicate major causal influences from upstream structural conditions, and dashed lines denote reinforcing feedback loops centred on limited policy effectiveness

Sorensen 2023). The magnitude and trajectory of weight gain depend not only on energy intake but also on individual properties, including genetic predisposition, metabolic state, and physiological regulation.

Conceptually, the bottom-up domain captures individual-level metabolic and behavioural feedbacks sustaining adiposity, whereas the top-down domain captures social and structural feedbacks that create and maintain obesogenic environments. At the physiological level, bottom-up processes involve metabolic, hormonal, and neuroendocrine regulation: hormonal dysregulation, including leptin and insulin resistance, reinforces appetite and energy storage, generating feedback loops that promote further weight gain (Table 2; Fig. 3).

Top-down regulation involves behavioural, social, and environmental drivers, including food availability, lifestyle patterns, and socioeconomic conditions (Kaisari et al. 2019; Sarvari et al. 2022). These factors shape energy intake, physical activity, and stress responses, while social feedbacks such as stigma and chronic stress further reinforce behavioural and metabolic patterns (Tables 2 and 3; Fig. 4). This framing emphasizes system-level feedbacks linking behaviour, physiology, and environment rather than simple, one-directional individual-level causation. These dynamics involve cross-scale feedback cascades, where metabolic, behavioural, and environmental processes reinforce one another, amplifying and stabilising the obese state.

Together, bottom-up and top-down processes form interacting feedback systems that stabilise the obese state and constrain transitions towards lower adiposity. In contrast to hypertrophic lakes, where system dynamics are dominated by a limited set of reinforcing feedbacks, regulatory processes in obesity remain distributed across interacting physiological, behavioural, and environmental domains. This configuration reflects a distinct form of bow-tie architecture, in which multiple feedback pathways remain active despite the concentration of control within a central regulatory core.

Although obesity is associated with substantial physiological and metabolic consequences, its functional effects are heterogeneous and not uniformly expressed as system-level dysfunction. Unlike hypertrophic lakes, which typically exhibit impaired ecosystem functioning, systemic overload in humans does not consistently produce overt dysfunction. This asymmetry reflects differences in regulatory architecture and is central to the comparison developed here.

Theories of alternative stable states and the set point of body weight

Transitions between alternative states are typically asymmetric: shifts towards hypertrophic phytoplankton dominance occur more readily than transitions back to oligotrophic or mesotrophic macrophyte-dominated conditions, as reinforcing feedbacks stabilize the turbid state and narrow the range of attainable system states, thereby increasing the threshold for a reverse regime shift (Scheffer and Carpenter 2003). These transitions emerge from interactions between slow pressures and threshold events that reorganise feedback structure and system dynamics (Fig. 5). This process can be described by distinguishing key components of regime shifts, namely slow drivers, triggering events, shifts in the regulatory core, and reinforcing feedbacks stabilising the new state, as summarised in Table 5.

Applying the alternative stable states framework to body weight regulation, two contrasting states can be distinguished: lean and obese. These states are maintained by stabilising physiological and behavioural feedbacks. Sustained positive energy balance can push the system beyond a threshold, triggering a transition to obesity that is stabilised by reinforcing feedbacks. As in lake ecosystems, these transitions are asymmetric, because feedback mechanisms constrain shifts towards lower adiposity. For analytical purposes, both lake ecosystems and human metabolic regulation are represented as transitions between contrasting states (clear vs. turbid; lean vs. obese). This abstraction highlights threshold dynamics and hysteresis, while recognising that real systems may exhibit intermediate or heterogeneous configurations. The theory of alternative stable states has a

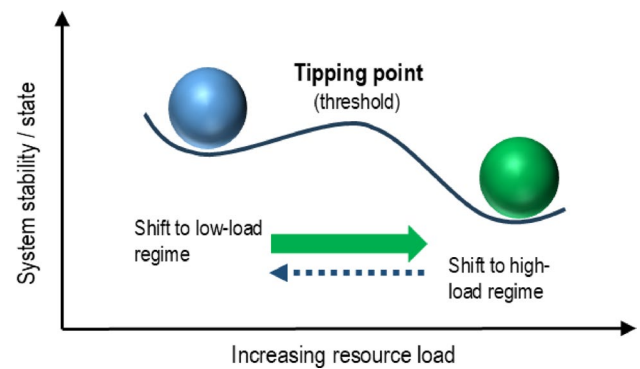


Fig. 5 Conceptual representation of regime shifts under increasing resource load. The stability landscape illustrates alternative stable states separated by a critical threshold: as resource input increases, the system may cross this threshold and shift from a low-load to a high-load state. Feedback-mediated stabilisation makes return to the original state contingent on substantially greater load reduction, reflecting hysteresis; examples of stabilising feedbacks and transition drivers are summarised in Tables 2, 3 and 4

Table 5 Drivers, triggers, and feedback-mediated regime shifts under resource overload in lakes and humans

System	Driver (slow pressure)	Trigger (threshold event)	Core regulatory shift	Feedbacks (post-shift)	Transition
Lakes	Nutrient loading (P, N)	Macrophyte collapse; sediment resuspension; hypoxia pulses	Biogeochemical saturation and oxygen depletion reducing stabilizing feedbacks	Internal loading; light limitation; cyanobacterial dominance; organic matter accumulation	Clear → turbid
Humans	Caloric surplus	Onset of leptin resistance; chronic stress; sleep disruption	Metabolic-hormonal dysregulation reducing regulatory sensitivity	Altered appetite feedbacks; reduced energy expenditure; adiposity persistence	Lean → obese

clear conceptual parallel in set-point models of body-weight regulation. These models propose that body mass is regulated around a relatively stable range maintained by physiological feedbacks, which resist sustained deviations and contribute to relapse after weight loss (Keesey and Hirvonen 1997; Speakman et al. 2011; Speakman and Hall 2023).

Contemporary formulations incorporate dynamic equilibrium and dual intervention points, while retaining the core idea that regulatory feedbacks stabilise body weight within a limited range (Speakman et al. 2011; Hall 2023).

Together, the theory of alternative stable states and models of body weight regulation indicate that different

regulatory systems can generate similar dynamical patterns, including stability, threshold behaviour, and hysteresis. This parallel is intended as a structural heuristic rather than a direct mapping between ecosystem equilibria and physiological regulation (Olson et al. 2019).

The alternative stable states framework has seen little explicit application to human obesity. In humans, obesity may be more appropriately interpreted as a shift in regulatory parameters (e.g. thresholds or feedback gain) within a dynamic equilibrium rather than as a transition between discrete potential wells; however, the framework remains useful for capturing threshold dynamics and hysteresis.

Remedial actions – restoration and treatment

Restoration of degraded hypertrophic lakes primarily targets the reduction of phosphorus and nitrogen availability (Table 6). Measures such as oxygenation, sediment removal, phosphorus inactivation, and biomanipulation aim to weaken reinforcing feedbacks, promote macrophyte recovery, and re-establish trophic control (Shapiro et al. 1975; Moss et al. 1996; Jeppesen et al. 2007; Tammeorg et al. 2024).

However, restoration is often slow, costly, and only partially effective. Feedbacks maintaining the hypertrophic state increase the threshold for reverse regime shifts, making long-term recovery uncertain; interventions rarely restore the original state and may introduce additional disturbances (Søndergaard et al. 2007). This reflects a general constraint imposed by feedback architecture: recovery is limited when reinforcing feedbacks stabilise high-load states and control is distributed across multiple components. Accordingly, both lake restoration and obesity treatment require multi-pronged interventions targeting different elements of the feedback network (Table 6).

Table 6 Restoration of hypertrophic lakes and treatment of human obesity

Lake	Human
1. Input reduction strategies: nutrient load control; wastewater treatment and management	1. Energy intake control: dietary modification, including reduced caloric intake; lower fat and sugar consumption, and increased fibre intake
2. Environmental and system management: catchment and shoreline buffering to reduce nutrient load	2. Behavioural interventions: lifestyle modification, including dietary habits, physical activity, and behavioural therapy
3. Direct system interventions: sediment removal; phosphorus inactivation; water oxygenation; thermal destratification; biomanipulation	3. Medical and clinical interventions: pharmacotherapy; bariatric surgery; psychological support

In humans, dietary, behavioural, pharmacological, and surgical interventions aim to reduce energy intake or alter components of the regulatory system, but long-term weight loss is difficult to sustain. Physiological and behavioural feedbacks resist change and promote relapse, consistent with models of body weight regulation (Wilborn et al. 2005; Speakman et al. 2011; Sorensen 2023).

These limitations reflect system-specific constraints on recovery. In lakes, recovery is primarily limited by external nutrient legacies and internal loading, whereas in obesity it is shaped by intrinsic regulatory features, including physiological set-points and behavioural feedbacks. In both systems, single interventions are often insufficient, as feedbacks are distributed around a capacity-limited regulatory core, allowing high-load states to persist and limiting sustained recovery.

Conceptual synthesis: a bow-tie framework for cross-system dynamics

Regulatory organization in complex biological systems can be described by a bow-tie architecture (also referred to as an hourglass architecture in systems biology), in which diverse inputs are funnelled through a central core that integrates mass, energy, and information and generates a wide range of outputs (Csete and Doyle 2004; Kitano 2004). While this structure supports efficient and robust functioning, it also creates structural vulnerability. Sustained overload of the core can saturate processing capacity, concentrate control within a limited set of pathways, and amplify positive feedbacks, reducing compensatory responses and allowing perturbations to propagate system-wide, thereby stabilising high-load states.

Within this framework, chronic overload can reorganise feedback structure and generate transitions into alternative stable states (Beisner et al. 2003; Scheffer and Carpenter 2003). These dynamics involve cascade-like propagation of effects across system components, although the structure of such cascades differs between systems. In lakes, they include trophic cascades across discrete trophic levels, whereas in obesity they take the form of cross-domain feedback cascades linking physiological, behavioural, and environmental processes.

The comparison between lake eutrophication and human obesity highlights a cross-domain alignment of key system components, linking state variables (trophic status vs. adiposity), control parameters (nutrient loading vs. caloric surplus), and feedback mechanisms (biogeochemical vs. metabolic–hormonal). This alignment provides a basis for comparing system dynamics without assuming direct equivalence between underlying mechanisms.

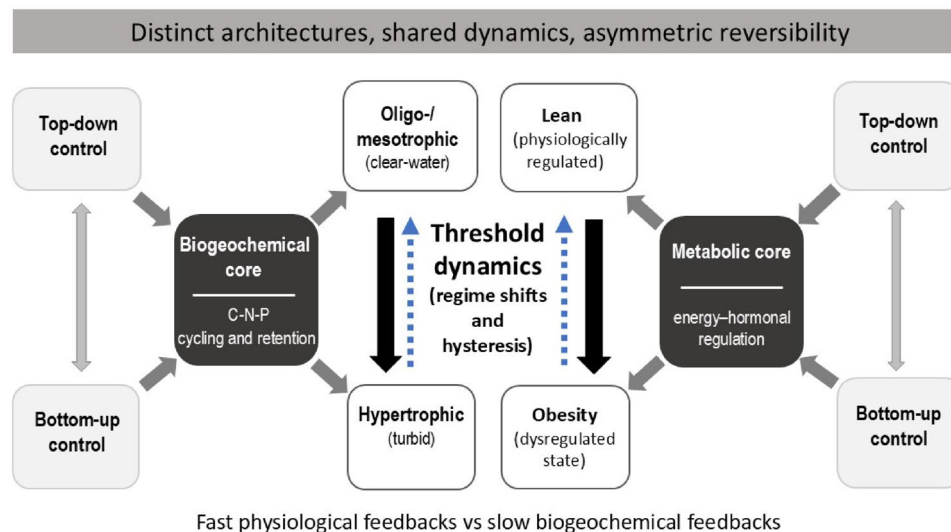


Fig. 6 Conceptual synthesis of bow-tie regulatory architecture. Distinct mechanisms in lake ecosystems and human metabolic regulation produce similar system-level dynamics under sustained resource overload. Diverse inputs are funnelled through central regulatory cores (biogeochemical vs. metabolic-hormonal), shaping system trajectory

ries and stability domains. Arrows indicate regulatory interactions and feedback pathways; regime shifts arise from threshold responses and reorganisation of feedbacks that stabilise alternative states. Two bow-tie diagrams provide a conceptual synthesis by explicitly linking regulatory architecture to system-level dynamics across systems

In lakes, diverse nutrient inputs and internal biogeochemical processes converge on a core controlling nutrient retention and oxygen dynamics, with oligotrophic conditions typically corresponding to macrophyte-dominated clear-water systems and hypertrophic conditions to phytoplankton-dominated turbid systems (Kornijów 2024).

In humans, environmental and behavioural inputs are integrated by a physiological core of metabolic-hormonal regulation (Fig. 6). The bow-tie representation provides a conceptual synthesis of this comparison by explicitly linking regulatory architecture to system-level dynamics.

Both systems exhibit hysteresis, yet their trajectories reflect contrasting feedback dynamics (fast physiological versus slow biogeochemical), resulting in asymmetric reversibility. In lakes, dynamics are primarily driven by external nutrient legacies and internal loading, whereas in obesity they are additionally shaped by physiological set-point regulation and behavioural-social feedbacks. This asymmetry implies that resilience is not a uniform property but depends on which variables are regulated and against which perturbations.

Finally, similar feedback-dominated states can arise from differently structured regulatory hierarchies: in lakes, dynamics are largely governed by bottom-up processes, whereas in obesity they are more tightly coupled to higher-level top-down drivers. This comparison illustrates how similar dynamical patterns can emerge from distinct regulatory architectures.

Implications for management and policy

Single-target interventions often fail because feedbacks are distributed around a central regulatory core. In both eutrophication and obesity, reinforcing feedbacks cluster around this core; interventions that address only one pathway may leave others intact, allowing the system to persist in a high-load state or return to it following disturbance. Effective management therefore requires coordinated actions targeting multiple components of the feedback architecture.

This perspective complements existing environmental management frameworks such as DPSIR and DAPSI(W) R(M) by shifting attention from pressures alone to the organization of regulatory processes. Interventions can thus be evaluated not only in terms of reducing external inputs, but also in terms of their capacity to modify feedback structure and restore regulatory flexibility.

From this viewpoint, effective strategies should aim to weaken reinforcing feedbacks, restore alternative pathways, and increase system responsiveness. This implies combining measures that act on external drivers with those that modify internal regulation and behavioural or structural processes, thereby addressing different components of the regulatory architecture.

The distinction between bottom-up and top-down processes is used analogically across systems: in lakes, it refers to trophic and resource-driven regulation, whereas in humans it encompasses physiological, behavioural, and socio-environmental influences.

Conclusions

This study shows that sustained resource overload acting on central regulatory cores can generate similar dynamical patterns across distinct biological systems. Within a bow-tie framework, eutrophication in lakes and obesity in humans can be understood as manifestations of feedback-driven dynamics, including alternative stable states, hysteresis, and asymmetric transitions.

Similar system-level behaviour arises from structurally different regulatory architectures, and resilience depends on system-specific organisation rather than being a uniform property. This perspective distinguishes general principles of complex system dynamics from system-specific mechanisms and highlights regulatory architecture as a key object of theoretical and empirical inquiry.

These results indicate that shared dynamics emerge not from common mechanisms, but from how regulatory architecture structures system responses to sustained resource overload. These differences imply that systems with more distributed regulatory architectures should exhibit slower but more persistent constraints on recovery, whereas systems with compressed feedback structures should exhibit sharper but more externally constrained reversibility.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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